

Nutritionally Induced Diabetes in Desert Rodents as Models of Type 2 Diabetes: *Acomys cahirinus* (Spiny Mice) and *Psammomys obesus* (Desert Gerbil)

Eleazar Shafir, Ehud Ziv, and Rony Kalman

Abstract

The dietary effects of hyperglycemia increasingly result in type 2 diabetes in humans. Two species, the spiny mice (*Acomys cahirinus*) and the desert gerbil (*Psammomys obesus*), which have different metabolic responses to such effects, are discussed. Spiny mice exemplify a pathway that leads to diabetes without marked insulin resistance due to low supply of insulin on abundant nutrition, possibly characteristic of a desert animal. They respond with obesity and glucose intolerance, β -cell hyperplasia, and hypertrophy on a standard rodent diet supplemented with fat-rich seeds. The accompanying hyperglycemia and hyperinsulinemia are mild and intermittent but after a few months, the enlarged pancreatic islets suddenly collapse, resulting in loss of insulin and ketosis. Glucose and other secretagogues produce only limited insulin release in vivo and in vitro, pointing to the inherent disability of the β -cells to respond with proper insulin secretion despite their ample insulin content. On a 50% sucrose diet there is marked lipogenesis with hyperlipidemia without obesity or diabetes, although β -cell hypertrophy is evident.

P. obesus is characterized by muscle insulin resistance and the inability of insulin to activate the insulin signaling on a high-energy (HE) diet. Insulin resistance imposes a vicious cycle of hyperglycemia and compensatory hyperinsulinemia, leading to β -cell failure and increased secretion of proinsulin. Ultrastructural studies reveal gradual disappearance of β -cell glucokinase, GLUT 2 transporter, and insulin, followed by apoptosis of β -cells. Studies using the non-insulin-resistant HE diet-fed animals maintained as a control group are discussed. The insulin resistance that is evident to date in the normoglycemic state on a low-energy diet indicates sparing of glucose fuel in muscles of a desert-adapted animal for the benefit of glucose obligatory tissues. Also discussed are the effect of *Psammomys* age on the diabetogenicity of the HE diet; the impaired function of several components of the insulin signal transduction pathway in muscles, which reduces the availability of GLUT4 transporter; the testing of several antidiabetic modalities for

the prevention of nutritional diabetes in *Psammomys*; and various complications related to the diabetic condition.

Key Words: β -cell apoptosis (*Psammomys*); high-energy diets; hyperinsulinemia; insulin resistance; insulin-resistant *Psammomys* protein kinase C ϵ ; insulin-signaling negative feedback; islet cell hypertrophy and disruption (*Acomys*); nutritional diabetes

Introduction

Investigation of propensity to hyperglycemia, insulin resistance, and diabetes in animals derived from the desert is of special importance in the age of increasing prevalence of type 2 diabetes, which has reached the extent of a world pandemic. Because the prevalence of type 2 diabetes is related to lifestyle and food intake, the nutritionally induced diabetes in rodents, which are free of any mutation, is of special interest as models of the impact of modern lifestyle and increased food consumption on the prevalence of diabetes (collusion of diabetes and obesity) that occurs in humans (Zimmet et al. 2001). In this article, we present two species, the spiny mice (*Acomys cahirinus*) and the desert gerbil (*Psammomys obesus*), which have different metabolic responses to the dietary impact. Their respective characteristics and the results of studies using each of these species are described below.

Spiny Mice

Spiny mice (*A. cahirinus* and related species) live in the arid areas of eastern Mediterranean countries and in North Africa. They are nocturnal large light brown mice that weigh 30 to 50 g and have fur bristles on their backs, hence the word “spiny” in their common name (Figure 1). They are born in litters of two to four after a gestation of 40 to 42 days, remarkably mature, covered with fur, and with open eyes.

The interest in spiny mice as a model of diabetes arose when overweight mice with glucosuria and ketosis were discovered by chance by Gonet and colleagues (1965) and Pictet and coworkers (1967) from the Albert Renold group in Geneva. The spiny mice were originally sent to Switzerland from Israel. After completion of a genetic study by Wahrman and Zahavi (1953), the mice were kept as pets in

Eleazar Shafir, Ph.D., M. Med Sc., and Ehud Ziv, Ph.D., are Professors of Biochemistry at the Diabetes Center, Hadassah University Hospital, and Hebrew University Hadassah Medical School, Jerusalem, Israel. Rony Kalman, D.V.M., Ph.D., DECLAM, is Director of the Authority for Animal Facilities at the Hebrew University of Jerusalem.

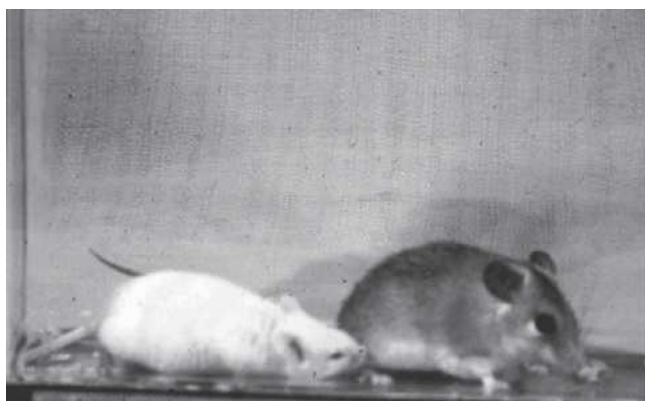


Figure 1 Spiny mouse (right) and albino mouse (left). Note the large size of the spiny mouse and the dark area of “spines” on its back.

bird cages in a private home and were maintained on bird food that comprised fat-rich pumpkin, sesame, and sunflower seeds. The mice were then transferred to the Institute of Biochemistry in Geneva and were fed a rodent chow supplemented by seeds, believed to enhance their fertility. In a series of studies, it was established that spiny mice manifest low insulin secretion capacity, low response to glucose, and faint first-phase insulin release, despite pancreatic islet hypertrophy and hyperplasia (Butter et al. 1980; Gonet et al. 1965; Gutzeit et al. 1974; Pictet et al. 1967; Stauffacher et al. 1970). Because up to 15 yr and about 40 generations had elapsed between the transfer of the wild mice from Israel to Switzerland and the discovery of diabetes, Renold and colleagues (1972) suggested that spiny mice might have undergone a mutation on the affluent nutrition in captivity, characterized by a defect in insulin release and compensatory islet replication.

Characteristics of Spiny Mice in the Geneva Colony

Nonketotic intermittent hyperglycemia was evident at the age of 9 to 12 mo in 10 to 20% of the animals in the colony, followed by a gradual or sudden lapse into ketosis with disintegration of the endocrine pancreas. Plasma insulin levels in the normoglycemic mice were not elevated and overt insulin resistance was not seen, apart from impaired glucose tolerance. The mice showed a low response to various secretagogues in the initial and late phases of insulin release, including glucose, arginine, glucagon, and cyclic AMP both in vivo and in vitro (Cameron et al. 1972; Junod et al. 1969; Rabinovitch et al. 1975). This response was attributed to several factors such as low islet adenylate cyclase activity, low cAMP response of β -cells after a glucose load, low amount of vincristine precipitable material in β -cell microtubuli through which insulin granules are extruded, and scarcity of autonomic islet innervation (Grill and Cerasi

1979; Gutzeit et al. 1974; Malaisse-Lagae et al. 1975; Orci et al. 1970). These findings were first interpreted as resulting from a genetic mutation in the animals maintained in captivity, but it is more likely that such responses of the islets may be a native characteristic of animals living in a desert environment, aimed at the protection of the pancreas against overstimulation. In later experiments, low insulin response could be potentiated by priming with high glucose concentration (Nesher et al. 1985). The possibility of amplifying insulin secretion from isolated islets indicates that the secretion defect is functional rather than inherent.

An electron-microscopic study revealed the presence of dense lysosome-rich bodies both in β - and α -cells (Orci et al. 1970), increasing the possibility that the excessive insulin storage of β -cell granules is contained by intracellular digestion. However, spiny mice at the stage of diabetes were hyperinsulinemic, and the prominent alterations were islet hypertrophy and proliferation, overdeveloped Golgi apparatus, hypergranulation, and increased insulin content with remarkable cell polymorphism (Gonet et al. 1965; Gutzeit et al. 1974; Pictet et al. 1967; Stauffacher et al. 1970).

Desert-collected Spiny Mice of the Jerusalem Colony

Spiny mice were collected from a site near the Dead Sea shore and a colony was established at the Hebrew University-Hadassah Medical School in Jerusalem to investigate whether the progression to diabetes during long-term affluent nutrition in captivity, may be evident also in newly collected animals. The spiny mice were fed a standard rodent chow on which they grew and multiplied well during >10 yr of observation. In contrast to the Geneva colony, no spontaneous hyperglycemia or glucosuria were seen in the approximately 300 mice (Gutman et al. 1972; Shafrir et al. 1972, 1974). To increase the nutrient challenge, the mice were placed on a purified carbohydrate-rich diet containing 50% sucrose and on a mixture of seeds containing 20% glycerolipids. The sucrose-rich regimen resulted in a remarkable induction of hepatic enzymes of glycolysis and lipogenesis leading to hyperlipidemia, mainly due to the elevation of very low density lipoprotein (VLDL¹) fraction (Gutman et al. 1972; Shafrir et al. 1974). The magnitude of enzyme induction and hyperlipidemia far exceeded that in spiny mice maintained on an isocaloric starch diet and was also more prominent compared with rats or albino mice that were fed a similar sucrose diet.

¹Abbreviations used in this article: DAG, diacylglycerol; DP, diabetes-prone; DR, diabetes-resistant; HE, high-energy; HGP, hepatic glucose production; IR, insulin receptor; IRS, insulin receptor substrate; LE, low-energy; PEPCK, phosphoenolpyruvate carboxykinase; PI-3K, phosphoinositol 3 kinase; PKB, protein kinase B; PKC, protein kinase C; TG, triglyceride; TGT, total glucose transport; TK, tyrosine kinase; TPA, phorbol ester.

The general metabolite and endocrine changes in spiny mice on different diets and the extensive induction of hepatic enzymes of glycolysis and lipogenesis are recapitulated and supplemented in Table 1. It should be remembered, however, that when spiny mice lose the β -cells and lapse into ketosis, they also lose weight and their liver enzyme pattern shows loss of glucokinase, phosphofructose kinase, and pyruvate kinase and an increase in glucose-6-phosphatase (Willms et al. 1970). The sucrose-rich diet resulted in substantial elevation of pancreatic insulin content and marked islet cell hyperplasia, but only a mild serum insulin increase and virtually no adipose tissue gain. In contrast, the fat-rich seed diet resulted in marked weight gain, impaired glucose tolerance, and moderate hyperinsulinemia associated with an increase in the pancreatic insulin content that exceeded those characteristics on a sucrose diet.

Comparison of Geneva and Jerusalem Spiny Mice Colonies

The mice and the diets of the Geneva and Jerusalem colonies were exchanged and fed ad libitum the Jerusalem standard chow or the Geneva diet, which was a rodent chow supplemented with seeds that contained up to 15% of fat in the diet. The Geneva diet caused a three-fold weight gain at 8 to 10 mo of age compared with albino mice on a rodent chow alone, and a two-fold gain compared with spiny mice fed the standard unsupplemented Jerusalem rodent diet (Shafrir et al. 1972). The hormonal, enzymatic, and metabolic patterns in the two colonies were similar, but one of the major differences was the adiposity on the Geneva diet. Insulin resistance preceded the hyperglycemia and was associated with mild hyperinsulinemia, progressing with time to more severe serum insulin elevation. Insulin resistance, which is associated with an insulin receptor defect as described in several obese animal species and obese humans was considered to be the cause of overt diabetes in the Geneva spiny mice maintained on the fat-rich seed-supplemented diet (Gutzeit et al. 1979). Indeed, a correla-

tion between insulin sensitivity and body fat content was demonstrated (Shafrir and Adler 1984). It is of interest that the propensity to obesity on the seed diet might be characteristic of the desert-derived spiny mice, either because of their lack of satiety in the desert or their particular taste for seeds. Laboratory albino mice fed a seed-supplemented diet gained only moderately in weight (Shafrir and Adler 1984). These comparative dietary studies in the two spiny mice colonies led to the conclusion that the low insulin response to glucose and other secretagogues as well as the lapse into diabetes comprise a trait of desert species exposed to nutritional plenty rather than a genetic aberration (Gutzeit et al. 1979).

Effect of Sucrose-rich Diet

On the sucrose-rich diet, spiny mice developed hepatomegaly, hyperactive lipogenesis, and gross VLDL elevation (Gutman et al. 1972; Shafrir and Adler 1984; Shafrir et al. 1974, 1975). Pancreatic islet size and insulin content increased (Figure 2), and ketosis or islet disintegration was apparent only at approximately 18 mo. The sucrose diet induced an elevation of circulating triiodothyronine (T3) (Shafrir 2000). Hepatic T4-T3 conversion was increased, whereas serum T4 levels tended to decrease. The activity of the T3-inducible hepatic mitochondrial flavin adenine dinucleotide glycerophosphate oxidase and K^+/Na^+ ATPase as well as body temperature were increased, indicating that the sucrose diet was associated with enhanced thermogenesis and energy-wasting metabolic cycling. The sucrose-rich diet most likely effected an adaptive defense mechanism, protecting against excessive weight gain and disruptive pancreatic islet lesion. After 18 mo of maintenance on sucrose-rich versus fat-rich diets, the number of spiny mice surviving was significantly higher on the sucrose diet, whereas on the fat diet a significant number of animals succumbed to islet cell disruption and diabetes. Islet insulin content, replication, and size were greater than on fat rich diet.

Table 1 Diet effects on metabolic parameters of spiny mice^{a,b}

Diet	Body weight (g)	3-mo gain (%)	Liver weight (g/100 g)	Fat tissue weight (g/100 g)	Serum triglyceride (mg/dL)	Serum cholesterol (mg/dL)	Serum glucose (mg/dL)	Serum insulin (mU/L)	Pancreas insulin (U/g)
Regular	44.0 $\pm$ 2.4	+5.0 $\pm$ 1.3	3.26 $\pm$ 0.12	3.35 $\pm$ 0.34	206 $\pm$ 16	118 $\pm$ 8	126 $\pm$ 4	34 $\pm$ 4	1.5 $\pm$ 0.2
Sucrose 50%	42.2 $\pm$ 1.7	-5.5 $\pm$ 1.7*	5.13 $\pm$ 0.24*	2.28 $\pm$ 0.40*	526 $\pm$ 52	282 $\pm$ 13*	134 $\pm$ 6	56 $\pm$ 8*	3.3 $\pm$ 0.3*
Fat (seeds)	58 $\pm$ 3.4	+29.3 $\pm$ 5.5*	3.14 $\pm$ 0.16	6.92 $\pm$ 0.45*	157 $\pm$ 12	156 $\pm$ 7*	173 $\pm$ 7*	84 $\pm$ 14*	2.0 $\pm$ 0.2

^aValues are means $\pm$ standard error for groups of 20 spiny mice maintained on diets for 3 mo. Asterisks denote a significant difference from mice on regular diet at $p < 0.05$ at least. Fat tissue was combined: epididymal, perirenal, mesenteric, and interscapular depots.

^bAdapted from Shafrir E. 1982. Intermediary metabolism during the development of obesity and diabetes in the desert rodent *Acomys cahirinus*. Int J. Obes 6:9-20.

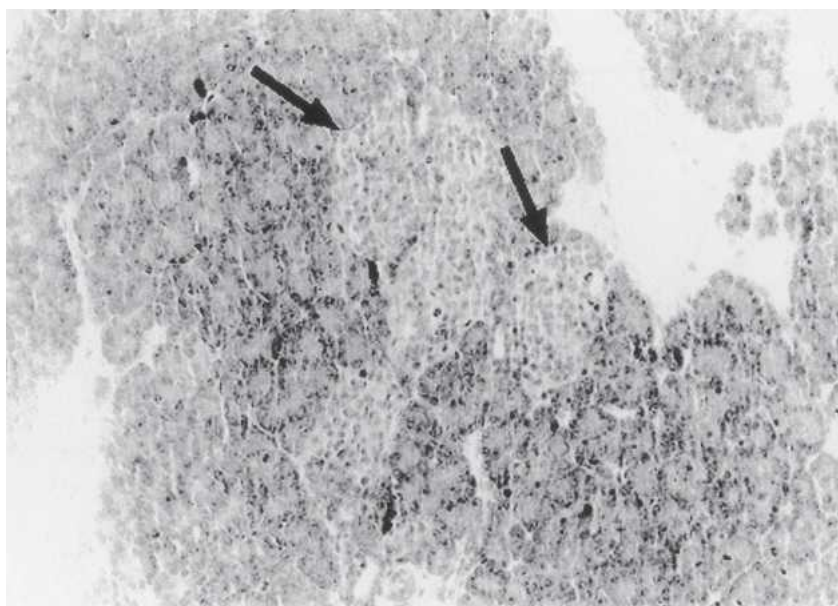


Figure 2 Hypertrophic and deformed islets of hyperglycemic spiny mice on a fat-rich diet.

The difference between spiny mice and other models of nutritionally induced diabetes is that spiny mice do not gradually progress to peripheral insulin resistance, hyperinsulinemia, hyperglycemia, and ketosis. Overnutrition primarily affects β -cells, causing hypertrophy and proliferation with propensity to disintegration (see Figure 2). This type of progression of diabetes differs from β -cell apoptosis caused by excessive insulin secretion pressure to compensate for the peripheral resistance, and it may represent another mode of nutritional diabetes development.

A species very close to *A. cahirinus* is *Acomys russatus* (golden spiny mice). *A. russatus* gains weight pronouncedly both on regular and fat-rich seed diets but does not exhibit pronounced hyperglycemia or hyperlipidemia, except increased hepatic triglyceride content, in association with high levels of circulating free fatty acids. Neither weight gain nor ketonuria was evident in *A. russatus* fed a sucrose-rich diet. On the fat diet, there was a smaller increase in activity of enzymes related to gluconeogenesis in *A. russatus* than in *A. cahirinus*, as well as a small suppression of glycolytic and lipogenic enzymes. Adipose tissue lipoprotein lipase activity increased in response to the fat-rich diet more markedly in *A. russatus* than in *A. cahirinus* in correlation with marked weight gain (Shafrir and Adler 1983).

Long-term maintenance of *A. cahirinus* on a high-sucrose diet had a deleterious influence on their reproduction and survival compared with those maintained on a regular diet. Pairs of spiny mice maintained on a 50% sucrose diet in two experiments lasting 8 and 18 mo gained less weight and exhibited a greater mortality rate of both parents and pups compared with mice kept on regular or fat-rich diets (Shafrir and Adler 1984). According to the number of pups born and the number of productive pairs,

the sucrose-fed mice were also less fertile. The litter size and the number of pups born per productive pair were slightly lower.

Finally, it should be mentioned that although the spiny mice of the genus *Acomys* have been classified as members of the rodent family of Murinae, other investigators (Agulnik and Silver 1996; Chevret et al. 1993) have reported evidence that these species are closer to Gerbilinae than to Murinae. Morphological characteristics and immunological studies have suggested a close relation to the Mongolian gerbil *Meriones unguiculatus*. Nevertheless, the *Acomys* species, which we have studied, were defined as Murinae by Aharoni (1932), and their chromosome pattern was determined by Wahrman and Zahavi (1953).

Concluding Remarks Related to *A. cahirinus* (Spiny Mice)

Spiny mice have been shown to gain weight markedly when maintained on a high-energy (HE¹) diet. Their weight gain is associated with a striking growth of pancreatic islets and with their insulin content. They do not readily respond to stimulation of insulin secretion. Diabetes occurs only in old animals, after spontaneous islet rupture that is accompanied by a loss of the rich insulin content.

Psammomys obesus

Diabetes in *P. obesus* was discovered by chance observation in desert rodents collected by the US Naval Medical Research Unit in Egypt in the 1960s. The animals were trapped on the sandy beaches of the Nile Delta and thought to be

rats. They were trivially nicknamed “sand rats,” which is a misnomer because instead of murines, they are gerbils that belong to the family Gerbillinae. They should be referred to as *Psammomys*, according to the classification of Thomas (1902, 1908).

The sand rats were sent to the laboratory of Schmidt-Nielsen of Duke University (Durham, NC). Investigators there were the first to report that diabetes occurs in the majority of *Psammomys* maintained on regular laboratory diet but not on a vegetable diet (Hackel et al. 1965a; Schmidt-Nielsen et al. 1964). The diabetes ranged from mild hyperglycemia with hyperinsulinemia to hypoinsulinemia with ketoacidosis, which was a terminal stage with short survival (Miki et al. 1966, 1967). These early investigations were performed mostly on the first generation of Egyptian *Psammomys* because attempts to establish a multigeneration colony were not successful due to low reproductive capacity on the regular chow. On this relatively high energy diet, a large proportion of the animals gained excess weight with a characteristic superscapular hump and became insulin resistant.

P. obesus of the Jerusalem Colony

Adler and colleagues (1976) collected *P. obesus* from the desert area north of the Dead Sea in Israel and successfully established a durable and reproducible colony in the Animal Facility of the Hebrew University-Hadassah Medical School (Figure 3). The animals were initially maintained on a “free choice” diet consisting of their desert staple of succulent leaves and branches of the salt bush *Atriplex halimus* collected from the Dead Sea region and fortified with a few pellets of regular rodent chow. During the more than 20 yr of the colony’s existence, three main groups were identified: normoglycemic-normoinsulinemic (A), normoglycemic-hyperinsulinemic (B), and hyperglycemic-hyperinsulinemic (C). The proportion of animals among these groups remained stable and predictable.

In an experiment that lasted 1 yr, during which >100 animals were removed at random from the colony, this classification of diabetes in *Psammomys* was made (Kalderon et al. 1986) (Figure 4). Approximately 32% of the animals were normoglycemic-normoinsulinemic and were referred



Figure 3 The gerbil *Psammomys obesus*.

to as Group A, and 26% were hyperinsulinemic but normoglycemic, with some gain in adipose tissue weight (group B). Group C was hyperglycemic despite the remarkable hyperinsulinemia. The very high insulin secretion in this group of *Psammomys* failed to promote peripheral glucose uptake, as determined by 2-deoxyglucose uptake. It also failed to restrain hepatic gluconeogenesis, as indicated by increased alanine conversion to glucose by isolated hepatocytes and the elevated activity of phosphoenolpyruvate carboxykinase (PEPCK¹) (Shafir and Ziv 1998). In group C, there was still a considerable deposition of adipose tissue and hypertriglyceridemia, demonstrating active hepatic lipogenesis and transport of lipoprotein-borne triglycerides to adipose tissue and overriding the insulin resistance of this tissue.

It is important to emphasize that the *Psammomys* liver is rich in lipogenic enzyme activity and is the main site of fat synthesis (Kalderon et al. 1983). The adipose tissue is rich in lipoprotein lipase and in the capacity to assimilate the preformed lipids (Chajek-Shaul et al. 1988) but poor in lipogenic enzymes compared with regular laboratory rats (Kalderon et al. 1983). It is also pertinent that a disproportionate increase of fatty acid-binding proteins was found in the liver of obese diabetic *Psammomys* (Lewandowski et al. 1997).

The last group (D) of *Psammomys* on a relatively high energy diet was hyperglycemic and inulinopenic and comprised only ~6% of the colony sample. All of these animals were lean, and their low plasma insulin levels indicated an exhaustion of insulin secretion. This group was on the verge of complete islet necrosis and ketonemia, which is a highly toxic condition in *Psammomys* because it is sensitive to ketoacidosis.

It should be stressed that the distribution in the colony described above does not necessarily represent gradual

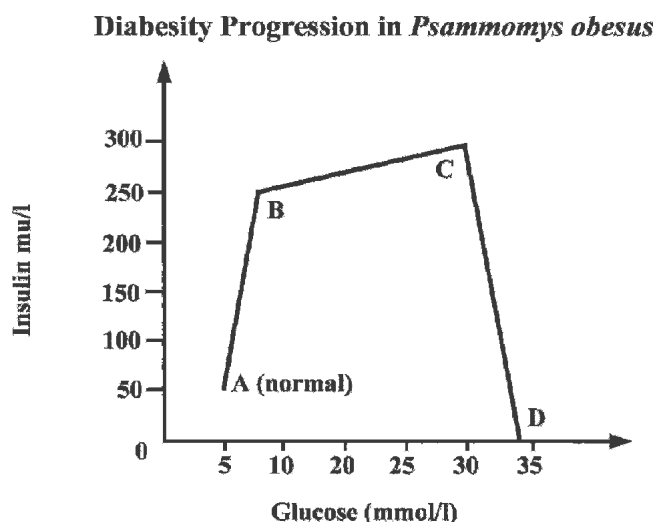


Figure 4 Stages of progression of *Psammomys obesus* to diabetes. Correlation between plasma insulin levels and blood glucose levels. Stage D signifies the collapse of the β -cell insulin secretion.

stages of diabetes progression from stages A to D. A longitudinal study of individuals in the *Psammomys* colony indicated that some animals may live for a long period of time as stage B, and some may directly lapse from stage A to C. Studies performed by Adler and coworkers (1988) have also revealed an inverted “U” shape of the curve of plasma insulin levels, in correlation with glucose levels. In these animals, a definite gradual and irreversible shift occurs from hyperinsulinemia with obesity to hypoinsulinemia with weight loss and fatal ketoacidosis. In *Psammomys*, the progression from stage A to C may be stopped or reversed by food limitation. Food restriction in stage C may prevent the lapse into ketosis and may restore insulin levels and normoglycemia, but there is no recovery from stage D.

A similar incidence of overt diabetes has been observed by Marquie and colleagues (1991) in *Psammomys* originating from Algeria. Approximately 40% of the animals developed the diabetes syndrome, with a few dying in ketosis. Aquichat Bouguerra and coworkers (2001, 2004) investigated the role of glucose and insulin on the synthesis of collagen in aortic smooth muscle cells of Algerian *Psammomys*. Earlier, Gernigon and colleagues (1994) had investigated the seasonal variations in the ultrastructure and production of testosterone-dependent proteins in the seminal vesicles of Algerian *Psammomys*. A branch of the Israeli *Psammomys* colony is bred in Australia (Barnett et al. 1994, 1995). It was observed that insulin resistance and hyperinsulinemia appear before weight gain, followed by adipose tissue accretion and diabetes. Tissue triglyceride (TG¹) deposition was driven by the ample hepatic lipogenesis, which continued unabated despite insulin resistance. With regard to the genetic characteristics, it is of interest that the product of a hypothalamic gene termed “beacon,” discovered by Collier and coworkers (2000) and Walder and colleagues (2002), was found to increase the food intake and body weight after intracerebroventricular injection. It also induced a two-fold increase in hypothalamic neuropeptide Y expression. The Australian and Jerusalem investigators found that the progression of *Psammomys* to diabetes may be reversed by reducing the nutrition in stage C, before apoptosis and β -cell degranulation sets in. At present, the Jerusalem colony is maintained on a specially devised low-energy (LE¹) diet (Kalman et al. 1993).

Psammomys maintained on a prolonged HE diet undergo massive β -cell degranulation, loss of insulin immunostaining, apoptosis, and necrosis. Like and Miki (1967) demonstrated stages of β -cell degranulation associated with glycogen deposition. The excessive glycogen in the islets most likely stems from massive penetration of glucose but lack of its metabolism because of degeneration of β -cell organelles, although Like and Miki did not observe reduction of islet protein synthesis until the late stages. Jörns and coauthors (2002) reinvestigated the β -cell changes during the progression of *Psammomys* to diabetes subsequent to hyperglycemia on HE diet, involving a gradual loss of β -cell insulin, glucokinase, and GLUT2 transporter. After 3 wk on HE diet, it was reduced by 70 to 95% of the initial

value, in correlation with the increasing blood glucose level. Ultrastructurally, different signs of necrotic destruction of pancreatic β -cells (e.g., the pyknosis of nuclei and a massive vacuolization in the cytoplasm) were observed to be accompanied by swollen mitochondria and dilated cisternae of the Golgi complex and of the rough endoplasmic reticulum. When the pancreas was removed from animals at stage D, β -cells exhibited apoptosis and nuclear DNA fragmentation (Donath et al. 1999; Nesher et al. 1999; Shafrir et al. 1991).

There is no direct evidence for the involvement of gluco- or lipotoxicity in the β -cell lesion in *Psammomys*. An attempt to prevent the possible effect of advanced glycation end products or of nitrous oxide by including the glycation inhibitor aminoguanidine in the hyperglycemic incubation medium was not effective in protecting β -cells in *Psammomys*. However, Kaneto and colleagues (1996) reported that reducing sugars may trigger apoptosis in β -cells of streptozotocin diabetic rats by the oxidative stress of glycation products. In their hands, the antioxidant N-acetyl-L-cysteine and aminoguanidine inhibited apoptosis. We presume that there may be species differences in reaction to hyperglycemia. In *Psammomys*, the prompt damage of β -cell architecture is probably the result of exhaustion due to the hypersecretion pressure before the eventual glucotoxic effect.

Psammomys in stage C show increased plasma proinsulin levels, up to one half of the circulating aminoassayable total insulin (Gadot et al. 1994). The inordinate secretion pressure may cause a swift exocytosis of immature insulin granules escaping before the C peptide cleavage. This evidence indicates that the compensation of the delayed glucose removal and suppression of gluconeogenesis by insulin were not effective because proinsulin has only a minute fraction of insulin activity. However, the high level of circulating proinsulin does not mean that its secretion is similar to that of insulin because the half-life of proinsulin is much longer than that of insulin (Glauber et al. 1986).

Selection of Defined Lines of *Psammomys*

As described above, the consumption of the HE diet was required for diabetes to be expressed in the animals we studied. The development of hyperglycemia is fast (7–14 days). In the original colony, the reaction of randomly chosen individual animals to the same HE diet may differ and there are always a few animals that remain normoglycemic even on the HE diet.

We used the HE diet to examine the diabetic potential of each individual and to identify the diabetic and nondiabetic margins of the population. By using an assortative mating system based on a minimal inbreeding method (Baker 1979), it was possible to separate the animals in the colony into two distinct lines, diabetes-prone (DP¹) and diabetes-resistant (DR¹), which differed phenotypically and genotypically (Kalman et al. 1993). Animals to be mated were chosen according to phenotypic parameters (postprandial

blood glucose and plasma insulin levels). The composition and digestibility of the diet are recorded in Table 2.

Interestingly, the existence of DR *Psammomys* had not been discovered earlier. Rice and Robertson (1980) kept a *Psammomys* colony in a general animal house on regular diet for several years and noted that they did not become diabetic. Apparently the DP animals died and only the DR survived. Their inability to produce diabetes in these animals led them to suggest a re-evaluation of the sand rat as a diabetes model.

Age-related Diabetogenicity

Psammomys fed the LE diet from weaning and transferred at different ages to the HE diet demonstrated that sensitivity to the development of diabetes increases from weaning, is greatest at 5 mo of age, and decreases thereafter (Ziv et al. 1999). At 5 mo of age, the obesity factor measured as the proportion of epididymal fat weight to total body weight is greatest, and the proportion of other organs to total body weight remains unchanged. In animals older than 7 to 8 mo, the potential to develop diabetes and obesity decreases. These changes are in correlation with the decrease in fertility in *Psammomys* from both DP and DR lines.

Metabolic and Reproductive Efficiency

The essence of thrifty metabolism is high metabolic efficiency that enables existence in an environment characterized by constant supply of LE diets. The artificial laboratory condition of an ad libitum-accessible HE diet creates a continuous input of energy and leads to hyperinsulinemia and hyperglycemia. The different sensitivity to the development of diabetes between the DP and the DR lines can be caused by one or more factors: differences in food intake, in hepatic and peripheral resistance, in pancreas activity, or in metabolic efficiency. To measure the metabolic efficiency, we followed animals during the period of their most rapid growth after weaning (2.5–3.0 g/day in males and 2.4–2.9

g/day in females) and calculated metabolic efficiency as the relation between digestible energy intake and weight increment (Kalman et al. 1993). There is no hyperphagia in either DP or DR lines when fed HE or LE diets; hyperglycemia is related to energy availability in the diet. The quantity of feces was significantly greater in animals fed the LE diet compared with animals fed the HE diet, in line with the carbohydrate digestibility in the diet. The metabolic efficiency in DP line *Psammomys* fed all diets was 6.0 to 6.6 Kcal/g of weight increase, whereas in the DR line, metabolic efficiency was 9.0 to 9.6 Kcal/g of weight increase.

In other studies performed on a branch of the Israeli colony in Australia (Barnett et al. 1995), hyperphagia was reported in the diabetic state of *Psammomys* fed a standard rat maintenance diet, but the weight gain did not exceed 15% of body weight. Differences between DP and DR lines are the result not only of their dietary-induced diabetes but also of other zootechnical and reproductive characteristics (Kalman et al. 1996). Reproductive efficiency in *Psammomys* is low compared with outbred rat strains (Baker 1979). The average number of weaned animals per female per week is 0.28 versus 1.0 to 1.5 in outbred rats (Weihe 1987). Reproductive efficiency is greater in the DP line compared with DR line females, due to the difference between non-reproductive females (22% in the DP line vs. 41% in the DR line) and the difference in the average number of births per female (3.3 in the DP line vs. 1.3 in the DR line). These two parameters create a difference in the total number of newborns per female during its reproductive life (11.5 newborn per female in the DP line vs. 5.4 in the DR line). No difference was observed in the average number of newborns per birth (2.8 vs. 2.7, respectively).

The period of most rapid growth of DP *Psammomys* on the LE diet is up to 65 days (2.5–3.0 g/day in males and 2.4–2.9 g/day in females). Growth continues up to 180 days but at a slower rate (1.4 and 0.8 g/day, respectively). The average weight of males is 264 ± 5 g and of females, 224 ± 7 g. The relative weight of most organs remains unchanged (Kalman et al. 1996). Adrenal glands are the only organ for

Table 2 Composition, energy, and digestibility of *Psammomys* diets^a

Components	HE diet		LE diet		Salt bush	
	%	Digestible portion	%	Digestible portion	%	Digestible portion
Protein	23.6	0.73	16.7	0.68	23.3	0.80
Fat	2.4	0.60	3.1	0.81	2.9	0.65
Carbohydrates	68.0	0.79	70.0	0.58	53.3	0.65
Ash	6.0	0.26	10.2	0.43	20.5	0.42
Total digestibility		0.76		0.62		0.69
Total energy (Kcal/g)	3.88		3.74		3.32	
Digestible energy (Kcal/g)		2.95		2.32		2.29

^aHE, high-energy; LE, low-energy; salt bush is native *Psammomys* food staple collected from desert.

which relative weight is significantly different between males and females in all age groups ($p < 0.001$), and it is always higher compared with albino rats (Kalman et al. 1996).

Inborn Insulin Resistance

Insulin resistance in *Psammomys* is an inherent innate characteristic even in the normoglycemic-normoinsulinemic stage A. In experiments in which plasma insulin was elevated by subcutaneous administration of exogenous bovine insulin in implants to normoglycemic-normoinsulinemic animals (stage A), only mild blood glucose reduction was observed compared with severe hypoglycemia in albino rats that received a similar dosage of insulin (Ziv et al. 1996). Bovine insulin was fully active in *Psammomys* as ascertained by the fact that the same insulin produced a marked reduction in serum glucose and TG levels in insulin-deficient stage D *Psammomys* (Ziv and Kalman 2000). Exogenous bovine insulin in the form of subcutaneous implants that release 2U/24 hr insulin for 10 days was introduced into the scurf of the animals in stage D *Psammomys* (Ziv et al. 1996). However, the exogenous insulin was not capable of lowering blood glucose levels in the HE diet fed highly insulin-resistant *Psammomys* of group C.

Our premise of liver and muscle primary insulin resistance as a species characteristic of *Psammomys* was confirmed by hyperinsulinemic-euglycemic clamp studies (Ziv et al. 1996) (Table 3). Quantitative data obtained on hepatic glucose production (HGP¹) and total glucose transport (TGT¹) indicate that insulin infusion did decrease the HGP and increase the TGT, again demonstrating the effectiveness of exogenous insulin. However, HGP was only par-

tially reduced (from 10.0 ± 0.6 to 3.8 ± 0.4 mg/min.kg), whereas in albino rats under the same conditions, the HGP was completely abolished (from 11.0 ± 0.5 to 0.7 ± 0.3 mg/min.kg). Lack of complete suppression of HGP in *Psammomys* was evident at a higher and longer lasting level of hyperinsulinemia than in rats. In addition, the limited elevation of TGT in the hyperinsulinemic clamped *Psammomys* (TGT *Psammomys* = 16.0 ± 1.9 mg/min.kg, whereas TGT albino rats = 39.0 ± 1.9 mg/min.kg) attests to the fact that the peripheral glucose utilization is low enough to be compensated by gluconeogenesis, avoiding lapse into hypoglycemia during exogenous insulin treatment. Insulin also failed to suppress in stage A *Psammomys* the activity of hepatic PEPCK, the rate-limiting enzyme of gluconeogenesis (Shafirir 1988), as well as the hepatic glucose output (Table 3).

The physiological importance of the data described above may be summarized as follows. We can assume that the inborn resistance in a desert animal imparts the capacity to direct the absorbed or endogenous scarcely available glucose for the benefit of glucose obligatory tissues, rather than to the muscle, which can utilize other sources of energy.

Mechanism of Insulin Resistance and Tyrosine Kinase Attenuation in *Psammomys*

To investigate the development of insulin resistance, the activity of tyrosine kinase (TK¹), the initiator of an insulin-signaling pathway, was studied in the liver and muscle of *Psammomys*. Kanety and colleagues (1994) found a low insulin receptor (IR¹) content in muscle and liver that is about one fifth that of the laboratory albino rat. However,

Table 3 Inborn insulin resistance in *Psammomys obesus*: Effect of exogenous insulin on metabolic characteristics compared with albino rats^a

	<i>Psammomys</i>		Albino rats	
	Without insulin	With insulin	Without insulin	With insulin
Serum glucose at sacrifice	4.8 ± 0.3	3.6 ± 0.5	5.7 ± 0.1	$2.5 \pm 0.1^*$
Serum insulin at sacrifice	510 ± 144	$1656 \pm 132^*$	150 ± 72	$1962 \pm 276^*$
Liver glycogen	71 ± 7	58 ± 7	48 ± 4	$27 \pm 3^*$
PEPCK ^c (mmol/min.mg protein)	320 ± 24	303 ± 6	222 ± 6	$124 \pm 11^*$
Clamp plasma glucose	4.9 ± 0.2		5.3 ± 0.2	
Clamp plasma insulin	1453 ± 132		834 ± 96	
Basal HGP ^c	10.0 ± 0.6		11.0 ± 0.5	
Insulin HGP	3.8 ± 0.4		$0.7 \pm 0.3^*$	
Total glucose transport	16.0 ± 1.9		$39.0 \pm 1.9^*$	

^aValues are means $\pm$ standard error for groups of 6 animals. Hyperinsulinemia was induced in *Psammomys* by subcutaneous implants releasing insulin for up to 60 days at a rate of 2 units per 24 hr. Hyperinsulinemic euglycemic clamp was performed in nonanesthetized chronically catheterized animals.

^bAdapted from Ziv E, Kalman R, Herskop K, Barash V, Shafirir E, Bar-On H. 1996. Insulin resistance in the NIDDM model *Psammomys obesus* in the normoglycemic, normoinsulinemic state. *Diabetologia* 39:1269-1275.

^cPEPCK, phosphoenolpyruvate carboxykinase; HGP, hepatic glucose production.

insulin binding and TK activity per receptor was normal both in vitro and in vivo. The TK activity was measured in progression to diabetes of stages B and C, as compared with the normoglycemic stage A. Basal phosphorylation of the isolated IR was comparable in these stages to that in the normoglycemic stage A, but the extent of TK activation by insulin was pronouncedly lower in stages B and C in liver and muscle (Kanety et al. 1994). The reduced activation by insulin was accompanied by a marked decrease in muscle GLUT4 protein and mRNA. (Shafrir and Ziv 1998). Both could be reversed by nutrition restriction to one half of their daily food intake for a few days. The recovery of TK activity was not complete but when hyperglycemia was corrected and insulin levels were reduced, the TK returned to normal range. These findings indicate that hyperinsulinemia and the related overexpression of protein kinase isoenzymes (see below) are the basic events responsible for deficient IR function causing insulin resistance, promoting multisite phosphorylation of the receptor, and most likely involving the inhibitory serine sites on the insulin signal transduction proteins (Shafrir 2001; Shafrir et al. 1999).

Overexpression of PKC ϵ —A Negative Feedback of Insulin Signal Transduction

Protein kinase C (PKC¹) in the gastrocnemius muscle of hyperinsulinemic *Psammomys* was found to be pronouncedly overexpressed (Ikeda et al. 2001; Shafrir et al. 1999). This enzyme group is now widely studied because of its preferential phosphorylation of serine and threonine residues on signaling pathway proteins, producing a negative feedback of this pathway. The PKC group includes at least 11 isoenzymes of which PKC was most pronouncedly overexpressed in *Psammomys* muscle. PKC ϵ was also translocated from the cytosol to muscle membrane to a larger extent than other PKC isoenzymes, which in addition to overexpression indicates an increased activity (Ikeda et al. 2001). Other PKC isoenzymes, particularly PKC α and PKC θ , also tended to be overexpressed.

The expression of PKC ϵ was compared in DR and DP *Psammomys* lines. PKC ϵ showed the highest overexpression in the skeletal muscle of *Psammomys* in the hyperglycemic-hyperinsulinemic stage C compared with the DR line. Significant overexpression of PKC ϵ in the normoglycemic stage A of DP *Psammomys*, compared with the DR line, indicates that PKC ϵ overexpression precedes the onset of overt hyperglycemia. This indication is in agreement with the observed innate insulin resistance referred to above. Thus, PKC ϵ overexpression in stage A may be considered as a marker of a “pre-diabetic” or “pre-insulinemic” stage and of the propensity of a given individual to progress to overt diabetes on affluent nutrition. It is, however, without untoward consequences as long as the diet is LE.

Because PKC ϵ overexpression resulted in impaired TK activation by insulin and reduced GLUT4 mRNA and protein, which indicates an impaired phosphoinositol 3 kinase

(PI-3K¹) activation, it was of interest to investigate whether PKC overexpression induces a further negative downstream effect on insulin signaling. The activity of protein kinase B (PKB¹)/Akt, an enzyme thought to be responsible for the activation of pleiotropic metabolic systems, was determined. The transfection of HEK 293 cells with IR and/or PKC ϵ plasmids, followed by stimulation with insulin or a phorbol ester (TPA), respectively, showed that the activation of PKC ϵ by TPA caused a reduction of PKB expression and inhibition of PKB activation (Ikeda et al. 2001). This result may be assumed to be caused by PKC-effected serine phosphorylation on insulin receptor substrate (IRS¹) on which the PKB and PI-3K functions depend.

The increased activity of PKC isoenzymes in muscle membrane, in IR proximity, suggested the involvement of PKC ϵ in the attenuation of IR/TK and IRS activation. Several PKC isoenzymes were shown to reduce the TK-catalyzed phosphorylation of the IR and IRS-1 (see Shafrir 2001). It was found that PKC ϵ overexpression was associated with reduced binding of insulin by muscle IR due to the reduction in the number of insulin receptors per cell. The downregulation of receptors was demonstrated in HEK 293 cells, which were transfected with human insulin receptors and PKC ϵ plasmids. Activation of the PKC ϵ by TPA reduced the amount of receptors to ~40% of the original number (Ikeda et al. 2001). This finding is in accord with observations of degradation of insulin receptors, induced by PKC ϵ and possibly other PKC isoforms. It is therefore likely that serine/threonine phosphorylation of the IR, and/or IRS-1, inhibits the TK activity via a feedback loop and is responsible for the deficient TK activation by insulin and IR degradation, leading to insulin resistance accentuation at stages B and C in *Psammomys* on the HE diet (Kanety et al. 1994).

PKC ϵ Overexpression and Muscle Lipid Content

The enhanced PKC ϵ activity and/or expression in *Psammomys* was found to be correlated with the increased muscle content of diacylglycerol (DAG¹) (Ikeda et al. 2001; Shafrir et al. 2002). DAG is an intermediate of both fatty acid esterification to TG and TG breakdown to fatty acids and glycerol. The increased muscle concentration of TG and DAG in *Psammomys* occurs in conditions of hyperinsulinemia and hyperglycemia, characteristic of stages B and C of *Psammomys*, and is most probably activating PKC ϵ . Among other PKC isoenzymes, PKC ϵ has been termed a “lipid second messenger” because it is dependent on intracellular DAG levels (Nishizuka 1995).

Psammomys as a Model for Testing Antihyperglycemic Drugs

Psammomys represent an excellent model to treat diabetes in stages B or C because the stages are reversible. Stage D,

in which the ϵ -cells are already compromised, is not suitable for treatment. Vanadyl sulfate was effectively used to prevent the hyperglycemia and hyperinsulinemia of *Psammomys* maintained on the HE diet (Shafrir et al. 2001). Administration of 5 mg/kg of vanadyl sulfate for 5 days resulted in a prolonged restoration of normoglycemia and normoinsulinemia as well as muscle GLUT4 transporter. Pretreatment with vanadyl sulfate significantly delayed the onset of hyperglycemia.

Rosiglitazone administered to *Psammomys* on the HE diet at 20 mg/kg for 2 wk also normalized the hyperglycemia and markedly reduced the hyperinsulinemia. Glucose metabolism in peripheral tissues was not the primary target of the beneficial effect of rosiglitazone, but rosiglitazone prevented the damage to pancreatic β -cells and loss of insulin, thus enabling insulin secretion to compensate for the peripheral insulin resistance (Hefetz et al. 2006).

Marquie and coworkers (1997) studied the oral agent S15261(3-[2-[2-[4-[2-[α -fluorenyl acetyl amino ethyl] enzyloxy] ethyl amino]1-methoxy ethyl] trifluoromethylbenzene), which also prevented the β -cell lesion in the HE diet-maintained *Psammomys*. Regranulation of β -cells and restored integrity of cytoarchitecture were observed.

Diabetes Complications in *P. obesus*

Hyperglycemia in *Psammomys* is associated with cataracts (Adler et al. 1984; Kohler and Knospe 1980), which appear after 2 to 4 mo on the HE chow. This association was also observed in other *Psammomys* colonies (Hackel et al. 1965a; Kuwabara and Okisaka 1976). Cataracts do not occur in the group A and in the normoglycemic-hyperinsulinemic group B of *Psammomys*. There is no evidence of retinal lesions. Another complication is due to microangiopathy, which is expressed by thickening of the intima and deposition of glycosaminoglycans observed in the Algerian *Psammomys* colony (Marquie et al. 1991). Degeneration of the intervertebral discs and spondylosis were noted in the Jerusalem colony (Silberberg et al. 1979). The hyperglycemic Jerusalem animals also showed evidence of neuropathy, which was inferred to be the result of a high pain threshold and a reduction in nerve conduction velocity (Wuarin-Bierman et al. 1987). Old *Psammomys* exhibited a tendency to develop hepatic malignancy (Ungar and Adler 1979).

With respect to the cataract appearance, it was observed that the level of galactokinase activity was very low in *Psammomys* red cells and most likely in ocular tissues, with or without cataracts. This evidence suggests that the formation of cataracts is precipitated by overeating and may act by conversion of glucose and galactose to sorbitol or galactitol, the accumulation of which is favored by hyperglycemia and by the content of galactose in the diet (Gutman et al. 1975). Cohen-Melamed and colleagues (1995) treated the *Psammomys* with acarbose, which resulted in a reduction not only of the profile of glucose but also of lens aldose

reductase activity, and which led to a significant preventive effect on cataract development. Borenshtein and coworkers (2001) observed that lipoic acid treatment was beneficial with regard to cataract development due to glucose lowering as well as to an increase in lens glutathione level.

Concluding Remarks Related to *Psammomys*

Psammomys in its native habitat is a healthy gerbil with a metabolic-endocrine system that is adjusted to desert life on a low caloric-density food, which enables reasonable survival. *Psammomys* is not hyperphagic; however, when high HE nutrition becomes available, it is predisposed to diabetes, obesity, and β -cell overtaxation. This predisposition is due to feedback inhibition of the insulin-signaling pathway, involving serine phosphorylation and stoppage of tissue glucose transport but not lipogenesis. This ill adaptation to nutrient excess is an outstanding example of the “thrifty gene” effect (Neel 1962; Wendorf and Goldfine 1991), and it therefore represents a suitable model for studying the mechanism of predisposition to insulin resistance and metabolic syndrome in human populations that evolve from scarcity to abundance in nutritional intake.

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